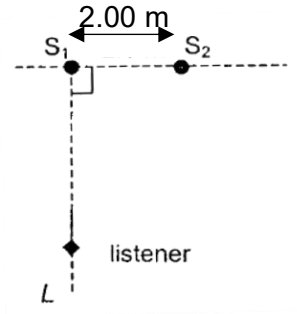


17. The figure below shows two speakers, S_1 and S_2 , placed 2.00 m apart. The two speakers are both part of a stereo system connected to the same source, which produces sound of wavelength 1.00 m.



What is the distance from S_1 that a listener must be, along the line S_1L , for him to hear a minimum in sound intensity?

- A** 1.20 m **B** 1.50 m **C** 2.00 m **D** 3.75 m

Ans: D

For the listener to hear a *minimum* in sound intensity, path difference between the two waves must be *half a wavelength*.

Let the distance, between the listener, L and S_1 be x .

By Pythagoras Theorem, distance, between the listener, L and S_2 is $\sqrt{(x^2 + 2^2)}$

This means that path difference between the two waves at L , $\Delta x = \sqrt{(x^2 + 2^2)} - x$

Therefore, we have $\sqrt{(x^2 + 2^2)} - x = \frac{\lambda}{2}$

$$\Rightarrow \sqrt{(x^2 + 2^2)} - x = \frac{1.0}{2}$$

$$\sqrt{(x^2 + 2^2)} = x + 0.5$$

$$x^2 + 2^2 = (x + 0.5)^2$$

$$x^2 + 2^2 = x^2 + x + 0.5^2$$

$$x = 2^2 - 0.5^2 = 3.75 \text{ m}$$

18 Two wires of the same material and length are connected in series in an electrical